Arun Kumar Boddapati

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https://github.com/arunbodd | https://arunbodd.github.io/Portfolio/
AIVA: https://chat.aivaportal.com/

Professional Summary

- **Strategic computational genomics leader** with 9 years driving AI-enabled precision medicine initiatives from research through clinical implementation.
- **Led cross-functional teams (3–7 FTEs)** across bioinformatics, data engineering, and translational science delivering $100K+ cost savings and 20–80% efficiency gains through AI-driven automation for Nextflow pipelines at CDC.
- **Principal-level technical leadership**: Architected and productionized 5+ pipelines (Nextflow) supporting public health surveillance, rare disease diagnostics, and whole genome/exome pipelines.
- **AI/ML innovation pioneer**: Designed LLM-based prompt engineering frameworks reducing pipeline development time by 80%; deployed RAG-enabled variant interpretation platform (**AIVA**) achieving 80.5% ACMG classification accuracy compared to BIAS (75.3%) and InterVar (60.65).
- **Translational impact**: Co-authored 15+ peer-reviewed publications in Cell, Immunity, and Science Advances.
- **Seeking Principal Scientist/Lead roles** to scale AI-driven automation successes to more complex portfolios and larger teams, expanding both technical and formal management responsibilities.

Leadership & Management

- **Team Leadership**: Cross-functional team coordination (across bioinformatics, data engineering, translational science), **matrix management**, **strategic initiative ownership**
- **Mentorship & Talent Development:** 1:1 coaching, technical skill development programs, performance feedback, junior scientist training and upskilling
- **Scientific Communication:** Presentation to **senior/executive leadership**, **stakeholder management**, cross-functional collaboration with data engineers and product managers.
- **Project & Resource Management:** Technical roadmap development, compute resource allocation, timeline/milestone tracking, best practice standardization
- **Strategic Planning:** Method development, contribution to departmental strategy and technical vision in AI implementation.

Technical Skills

- **AI foundations**: Prompt engineering, LLM-based workflow design, vector embeddings and semantic search, RAG architectures, MCP based agentic/tool-calling and evaluation of model performance on domain-specific tasks.
- **Multi-omics:** RNA-seq (bulk, Iso-Seq), Single-cell, Spatial, CITE-Seq, targeted proteomics (NULISA-Seq, Olink), WES/WGS, ChIP-seq, MeRIP-seq, metagenomics
- **Pipelines & infrastructure:** Nextflow, nf-core, Snakemake, WDL, Docker/Singularity, AWS, Azure, CI/CD, Slurm, PBS, SunGridEngine
- **Machine learning:** Supervised/unsupervised ML, deep learning (autoencoders, large language models), model validation and feature selection for biomarker discovery
- **Programming & analysis:** Python (pandas, numpy, scikit-learn, PyTorch), R (tidyverse, Bioconductor), Bash, SQLVisualization & reporting: R Shiny, RMarkdown, dashboarding and automated QC/reporting workflows

Professional Experience

Health Scientist (Technical Lead) | Booz Allen Hamilton (CDC OAMD-Data Modernization Accelerator)

March 2025 – January 2026 | Atlanta, GA

- **AI strategy and innovation**: Led the design of an **LLM-driven prompt engineering framework** to automate Nextflow pipeline development for multiple CDC programs, cutting development time by 20–80% (most for complex pipelines) and enabling program managers to replan roadmaps around faster delivery.
- **Cross functional Team leadership & Resource coordination:** Led coordination across bioinformatics, data engineering, and translational science teams (3-5 FTEs) to standardize best practices, maintain robust reproducibility across multiple CDC modernization projects, and align technical roadmaps with program milestones and stakeholder timelines.
- **Scalable Pipeline Development:** Optimized Nextflow-based pipelines (Cyclone, MycoSNP) reducing analysis turnaround by 20–50%; architected cloud-native (AWS) infrastructure supporting clinical studies with reproducible CI/CD workflows.

Bioinformatics Scientist (Senior Staff) | Leidos (CDC OAMD-Scientific Computing)

July 2022 – January 2025 | Atlanta, GA

- **Technical Mentorship & Pipeline Development:** Designed and optimized production grade Nextflow pipelines (Legionella, Tau-typing, Neisseria); mentored 3 junior bioinformatics analysts on pipeline development, machine learning validation, and reproducible workflows; built sustainable teams through monthly workshops on Nextflow pipeline engineering and cloud infrastructure.
- **Biomarker Discovery & Machine Learning**: Pioneered deep learning biomarker discovery for silicosis risk prediction using PBMC single-cell RNA-seq cohort (GLaeDOS/autoencoder), achieving 80% accuracy on risk stratification and supporting early-stage translational programs.
- **Mechanism of Action Analysis**: Integrated phylogenetic and molecular epidemiology workflows to support understanding of pathogenic mechanisms and therapeutic targets.

Senior Bioinformatics Analyst (Specialist IC) | Emory Primate Research Center

January 2022 – July 2022 | Atlanta, GA

- **Single-Cell RNA-Seq for Drug Discovery**: Led single-cell RNA-seq analyses for MVA vaccine and COVID-19 immunological studies in non-human primates; contributed actionable MOA insights informing therapeutic evaluation and go/no-go decisions.
- **Multi-Omics Integration**: Designed and implemented Snakemake AWS-optimized workflows integrating genomics and transcriptomics data; achieved $100K+ in cost savings through optimized QC and analysis strategy supporting translational research programs.

Bioinformatics Analyst (Specialist IC) | Emory Primate Research Center

June 2020 – December 2021 | Atlanta, GA

- **High-Impact Translational Study**: Led comprehensive RNA-seq bioinformatics for IMPACC study (5,000+ COVID-19 patients); optimized transcriptomics pipeline achieving 60% runtime reduction, directly supporting late-stage clinical translational programs.
- **Molecular Biology Expertise**: Gained deep understanding of PBMC RNA-seq library preparation, QC workflows, and clinical-grade reporting standards; collaborated with wet-lab teams on experimental design and data integration.

Bioinformatics Analyst II | Leidos Biomedical Research (NIAID Bioinformatics Core Support)

April 2018 – June 2020 | Bethesda, MD

- **Multi-Omics Analysis at Scale**: Supported 28+ genomics and transcriptomics projects integrating RNA-seq, microarray, and single-cell data for clinical and translational insights.
- **Peer-Reviewed MOA Research:** Co-authored Journal of Infectious Diseases study characterizing TIGIT+ CD8+ T cells in HIV; demonstrated immune mechanism and informed checkpoint-based cure strategies, exemplifying translational impact on drug discovery.
- **Automated QC & Reporting**: Developed Python and R Shiny applications for automated QC, visualization, and clinical reporting; optimized Slurm/PBS pipelines, saving 100+ hours and accelerating time-to-insight.

Volunteer Experience

Founding Bioinformatics Engineer | Mamidi Health

January 2026 – Present | Atlanta, GA

- **Led design and development of** AIVA, an AI-powered variant analysis and reporting platform that reduces end-to-end genetic diagnostics turnaround from weeks to minutes.
- **Oversaw design and execution of** **benchmarking** for AIVA versus BIAS-2015 and InterVar on ClinVar-classified variants, validating phenotype-aware ACMG interpretation gains.

Bioinformatics Consultant (Volunteer) | Cure VCP Disease Foundation / Dr. Pant Lab, Emory University

August 2024 – Present | Atlanta, GA

- **Targeted Proteomics Analysis**: Designed and executed NULISA-Seq targeted proteomics analysis for protein dysfunction research (published in HGG Advances, 2025).
- **Biomarker Discovery**: Developed R Shiny dashboard using limma linear modeling and leave-one-out cross-validation for target identification supporting rare disease research.

Education

- Master of Science, Bioinformatics | Indiana University, Indianapolis, IN | 2016–2017
- Master of Science, Biomedical Sciences | Symbiosis International University, India | 2012-2014
- Bachelor of Technology, Biotechnology | JNTU, India | 2007-2011

Publications

1. Dulski J, **Arun K. Boddapati**, et al. (2025). "Targeted plasma proteomics uncover proteins associated with KIF5A-linked SPG10 and ALS spectrum disorders." HGG Advances, 6(1)
2. Hoang TN, Pino M, **Arun K. Boddapati**, et al. (2021). "Baricitinib treatment resolves lower-airway macrophage inflammation and neutrophil recruitment in SARS-CoV-2-infected rhesus macaques." Cell, 184(5), 1415–1429.
3. Prasida Holla et al. (Including **Arun K. Boddapati**) (2020). "Shared transcriptional profiles of atypical B cells suggest common drivers of expansion and function in malaria, HIV, and autoimmunity." Science Advances, 7, eabf8457.
4. Jana et al. (including **Arun K. Boddapati**) (2019). "TIGIT+ CD8+ T cells retain strong intrinsic cytotoxic capacity and are restored with checkpoint blockade." The Journal of Infectious Diseases, 220(12), 1622–1630.
5. Routhu NK, et al. (including **Arun K. Boddapati).** (2021). "A modified vaccinia Ankara vector-based vaccine protects macaques from SARS-CoV-2 infection, immune pathology, and dysfunction in the lungs." Immunity, 54(12), 2859–2876.

Additional publications: **Google Scholar Profile**